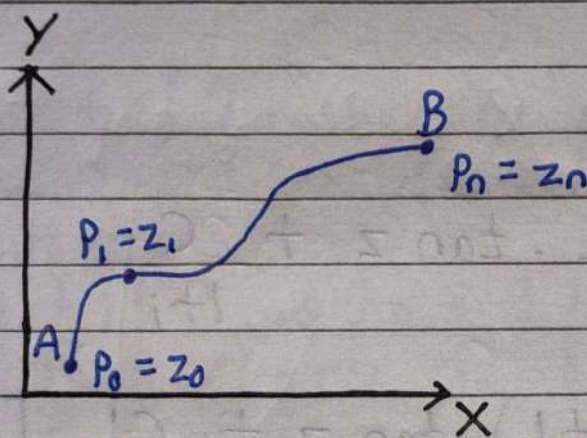


Line Integrals in Complex Plane

Let, $f(z)$ be a continuous function of the complex variable $z = x + iy$, defined at all points along a curve C with end-points A & B . Divide the curve C into n sub-arcs at the points:

$$A = P_0(z_0), P_1(z_1), P_2(z_2), \dots, P_n(z_n) = B$$



Let, z_{i-1} & z_i be the endpoints of the i -th sub-arc and define the change in z as:

$$\delta z_i = z_i - z_{i-1}$$

Let, ξ_i be any point chosen on the sub-arc $P_{i-1}P_i$, then the limit of the sum as $n \rightarrow \infty$ and $\delta z_i \rightarrow 0$ is defined as:

$$\sum_{i=1}^n f(\xi_i) \delta z_i, \text{ when } n \rightarrow \infty \text{ \& } \delta z_i \rightarrow 0.$$

If this limit exists, it is called the line integral of $f(z)$ along the curve C i.e. $\int_C f(z) dz$.

In case points P_0 & P_n coincide (i.e. $A=B$), then the curve C forms a closed curve. The integral along such a closed path is called a Contour Integral, which is denoted by:

$$\oint_C F(z) dz.$$

To evaluate the integral using real variables, substitute $F(z) = u + iv$ and $z = x + iy \Rightarrow dz = dx + i dy$.

$$\int_C F(z) dz = \int_C (u + iv)(dx + i dy)$$

$$= \int_C (u dx + i u dy + i v dx - v dy)$$

$$= \int_C (u dx - v dy) + i \int_C (v dx + u dy).$$

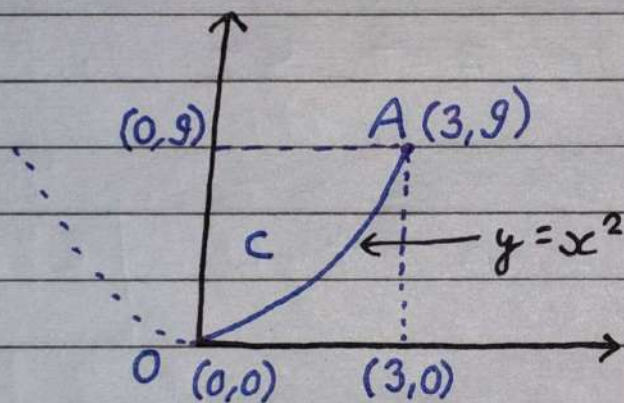
8. Find the value of the integral $\int_C \{ (x+y) dx + x^2 y dy \}$

- i. along $y = x^2$ having $(0,0)$ and $(3,9)$ as end points.
- ii. along $y = 3x$ between the same points.

Solⁿ:

Given, $\int_C \{ (x+y) dx + (x^2 y) dy \}$

- i. along $y = x^2$ joining $(0,0)$ & $(3,9)$,



$$y = x^2$$

$$\Rightarrow dy = 2x dx$$

$$\int_C \{ (x+y) dx + (x^2 y) dy \}$$

$$= \int_0^3 \{ (x+x^2) dx + (x^2 \cdot x^2) dx (2x) \}$$

$$= \int_0^3 (x + x^2 + 2x^5) dx$$

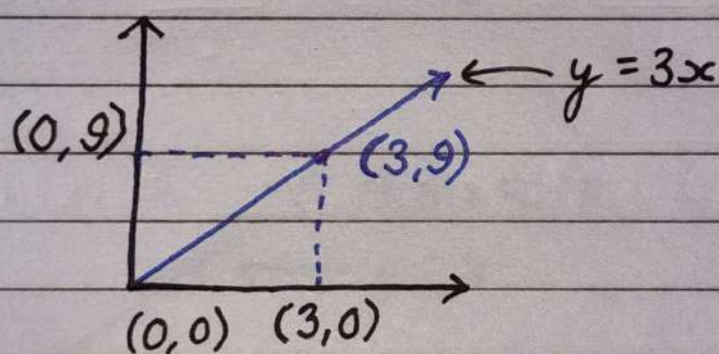
$$= \left[\frac{x^2}{2} + \frac{x^3}{3} + \frac{2x^6}{6^3} \right]_0^3$$

$$= \frac{9}{2} + \frac{27^3}{3} + \frac{729}{3}$$

$$= \frac{9}{2} + 9 + 243$$

$$= \frac{513}{2}$$

ii. along $y = 3x$ between $(0, 0)$ & $(3, 9)$,



$$y = 3x$$

$$\Rightarrow dy = 3 dx$$

$$\int_C \{ (x+y) dx + (x^2 y) dy \}$$

$$= \int_0^3 \{ (x+3x) dx + (x^2 \cdot 3x) (3 dx) \}$$

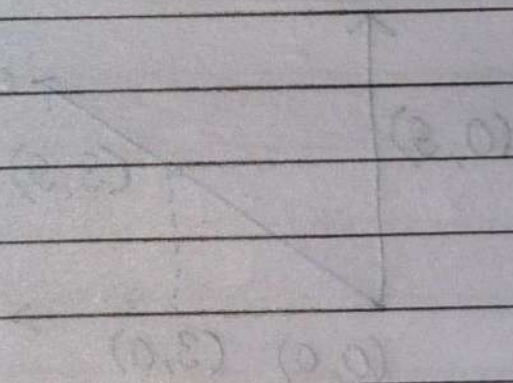
$$= \int_0^3 (4x + 9x^3) dx$$

$$= \left[\frac{4x^2}{2} + \frac{9x^4}{4} \right]_0^3$$

$$= \left[2x^2 + \frac{9x^4}{4} \right]_0^3$$

$$= 18 + \frac{729}{4}$$

$$= \frac{801}{4}$$



$$x=y$$

$$x^2=y$$

$$\int_0^2 (y - x^2) dx = \int_0^2 (x - x^2) dx$$

$$\left[\frac{1}{2}x^2 - \frac{1}{3}x^3 \right]_0^2 = \left[\frac{1}{2}(2)^2 - \frac{1}{3}(2)^3 \right] = \left[2 - \frac{8}{3} \right] = \frac{6}{3} - \frac{8}{3} = -\frac{2}{3}$$

$$z=0 \Rightarrow x=0 \text{ \& } y=0$$

$$z=1+i \Rightarrow x=1 \text{ \& } y=1$$

$$z = x + iy$$

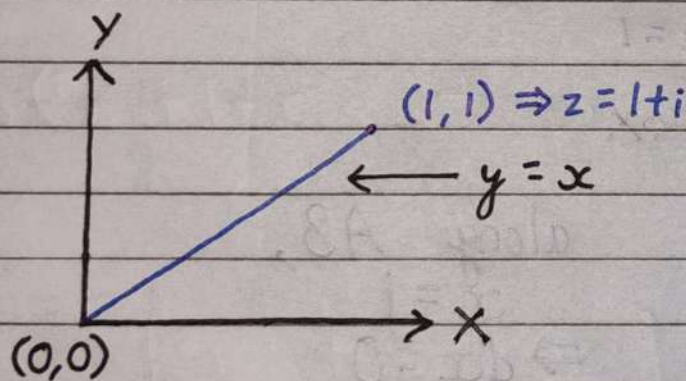
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8. Evaluate $\int_0^{(1+i)} (x-y+ix^2) dz$

- along the straight line from $z=0$ to $z=1+i$.
- along real axis from $z=0$ to $z=1$ and then along a line parallel to the imaginary axis from $z=1$ to $z=1+i$.
- along the imaginary axis from $z=0$ to $z=i$ and then along a line parallel to real axis from $z=i$ to $z=1+i$.

Solⁿ:

i.



$$y=x \Rightarrow dy = dx$$

$$dz = dx + i dy = dx + i dx$$

$$\int_0^{(1+i)} (x-y+ix^2) dz = \int_0^1 (x-x+ix^2) (dx+i dx)$$

$$= \int_0^1 ix^2 \cdot dx(1+i)$$

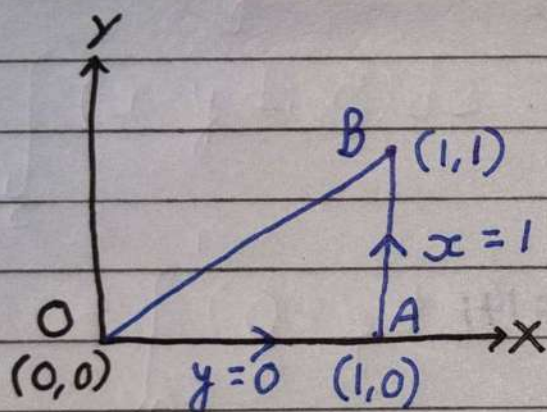
$$= (1+i) \int_0^1 ix^2 dx$$

$$= i(1+i) \left[\frac{x^3}{3} \right]_0^1$$

$$= (i+i^2) \cdot \frac{1}{3}$$

$$= \frac{i-1}{3} = -\frac{1}{3} + i \cdot \frac{1}{3}$$

ii.



along OA,

$$y=0$$

$$\Rightarrow dy=0$$

$$\Rightarrow dz = dx + idy = dx$$

along AB,

$$x=1$$

$$\Rightarrow dx=0$$

$$\Rightarrow dz = dx + idy = idy$$

$$\int_C (x-y+ix^2) dz$$

$$= \int_{OA} (x-y+ix^2) dz + \int_{AB} (x-y+ix^2) dz$$

$$= \int_0^1 (x - 0 + ix^2) dx + \int_0^1 (1 - y + i) i dy$$

$$= \left[\frac{x^2}{2} + i \frac{x^3}{3} \right]_0^1 + i \left[y - \frac{y^2}{2} + iy \right]_0^1$$

$$= \frac{1}{2} + i \cdot \frac{1}{3} + i \left(1 - \frac{1}{2} + i \right)$$

$$= \frac{1}{2} + i \cdot \frac{1}{3} + i - i \cdot \frac{1}{2} + i^2$$

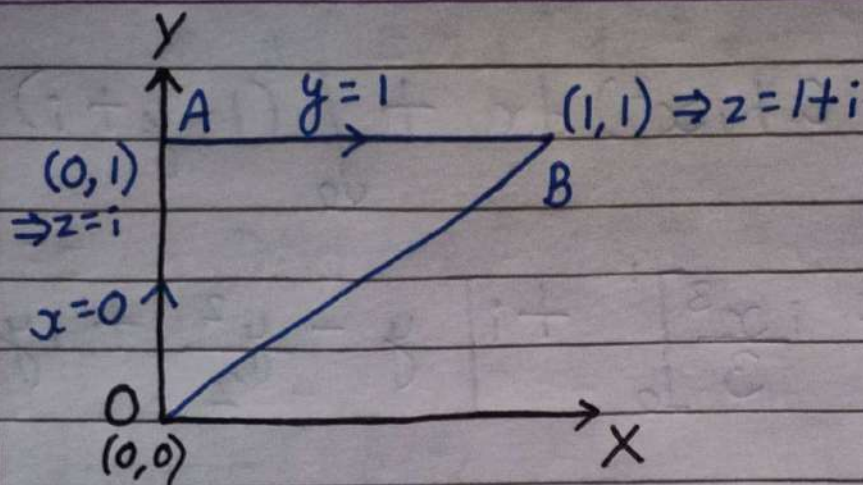
$$= \frac{1}{2} + \frac{i}{3} + i - \frac{i}{2} - 1$$

$$= -\frac{1}{2} + i \left(\frac{1}{3} + 1 - \frac{1}{2} \right)$$

$$= -\frac{1}{2} + i \left(\frac{2+6-3}{6} \right)$$

$$= -\frac{1}{2} + i \cdot \frac{5}{6}$$

iii.



along OA,
 $x=0$
 $\Rightarrow dx=0$
 $\Rightarrow dz=idy$

along AB,
 $y=1$
 $\Rightarrow dy=0$
 $\Rightarrow dz=dx$

$$\int_C (x-y+ix^2) dz$$

$$= \int_{OA}^1 (0-y+0) idy + \int_{OB}^1 (x-1+ix^2) dx$$

$$= i \left[-\frac{y^2}{2} \right]_0^1 + \left[\frac{x^2}{2} - x + i\frac{x^3}{3} \right]_0^1$$

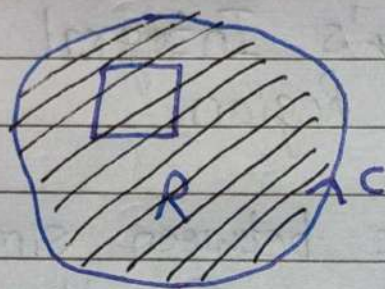
$$= -i \cdot \frac{1}{2} + \frac{1}{2} - 1 + i \cdot \frac{1}{3}$$

$$= -\frac{1}{2} - i \left(\frac{1}{2} - \frac{1}{3} \right)$$

$$= -\frac{1}{2} - i \cdot \frac{1}{6}$$

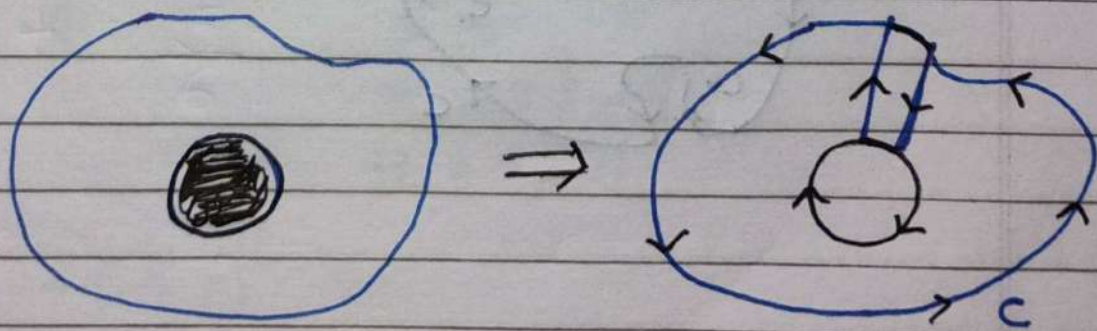
Simply Connected Region

A region is called simply connected if every closed curve in the region encloses points of the region fully - i.e. every closed curves lying in it can be contracted indefinitely without passing out of it.



Multiply Connected Region

A region is called multiply connected if every closed curve in the region encloses not the points of the region only i.e. every closed curve lying in it can be contracted indefinitely passing out of it.



Note: A simply connected region is converted into multiply connected region by making holes in it, & for vice-versa holes are given cut.